

Adnan Rizvi

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Profile

Software Engineer with hands-on experience in backend development, data engineering, and machine learning, actively working with Python, FastAPI, RESTful APIs, and Redis, with exposure to Generative AI workflows. Strong foundation in SQL, machine learning, and deep learning, supported by industry experience and the ability to build scalable, data-driven solutions in collaborative environments.

Skills

Programming Languages: Python, Java

AI / Machine Learning: Machine Learning, Deep Learning, Artificial Intelligence, Generative AI, RAG, CNN, NLP, TensorFlow, Scikit-learn

Backend & APIs: Flask, FastAPI, RESTful APIs, Postman, Redis

Data & Analytics: SQL, Tableau, Power BI, Excel

Web Technologies: HTML, CSS, JavaScript, Web Scraping (Scrapy)

Cloud & Tools: AWS, Git, GitHub, Snyk

Work Experience

Software Engineer (Trainee)

12/2025 – 04/2026

PeopleStrong — Goldman Sachs

- Built scalable services in Python–FastAPI delivering ML and GenAI capabilities.
- Integrated predictive and generative models with production-ready systems alongside data scientists.
- Enhanced client-side experience through optimized request–response handling.
- Mitigated GPL/LGPL risks by migrating dependencies to MIT/MPL/Apache-compliant using Snyk.
- Designed RESTful AI services and orchestration workflows using Langflow.
- Ensured reliability, maintainability, and scalability through clean architecture and engineering standards.

Data Science Intern

04/2025 – 11/2025

UTILIZED

- Built end-to-end Machine Learning pipelines involving data preprocessing, feature engineering, model training, evaluation, and performance optimization using Python and scikit-learn.
- Developed Deep Learning models using TensorFlow and Keras for classification and predictive analytics while experimenting with different architectures and hyperparameters.
- Performed Exploratory Data Analysis (EDA) using Pandas, NumPy, and Matplotlib, identifying trends and preparing clean datasets for model development.
- Wrote SQL queries for data extraction, transformation, and analysis, enabling data-driven insights and supporting business reporting.

Data Engineer Intern (Vocational Training)

08/2023 – 09/2023

Tata Steel Ltd.

- *Project: Integrated Supply Chain Management (Transformation Department)* — worked under the Transformation Department, analyzing operational workflows to improve supply chain efficiency.
- Designed and enhanced a vehicle check-in system, reducing entry time by 30% and improving operational process efficiency.
- Used SQL to analyze operational data, identify bottlenecks, and contribute to a 15% improvement in check-in workflow efficiency.
- Collaborated with cross-functional teams to support backend integration and data-driven process improvements.
- Generated reports and operational insights to support informed decision-making and smoother supply chain execution.

Projects

Crime Classification using Artificial Intelligence with Deep Learning | GitHub

- Developed a CNN-based model to classify crime scene images, reducing misclassification by 50% and achieving 84% accuracy.
- Enforced end-to-end pipeline including data preprocessing, model training, and evaluation using TensorFlow.
- Built an interactive web application using Flask, HTML, CSS, and JavaScript to enable user-friendly image uploads and predictions.

MCP Orchestration Framework with Langflow — PeopleStrong | GitHub

- Built a custom MCP server to orchestrate LLM tool interactions using Langflow, enabling deterministic routing of structured tools.
- Designed a modular Python-based orchestration pipeline integrating financial calculators (SI, CI, EMI) and external Weather APIs.
- Implemented secure AST-based math evaluation to ensure safe execution of user-generated expressions.
- Demonstrated LLM-tool orchestration workflows for reliable real-world task execution instead of free-form generation.
- **Tech Stack:** Python, Langflow, Model Context Protocol (MCP), FastAPI, REST APIs, AST Parsing, Tool Orchestration.

AI RAG System – Semantic PDF Question Answering | GitHub

- Built an end-to-end RAG system to query PDF documents and generate context-aware answers using LLMs.
- Implemented document chunking and semantic embeddings with LangChain and stored vectors in Pinecone for efficient similarity search.
- Integrated Gemini LLM for grounded response generation using retrieved document context.
- Developed a query rewriting pipeline to improve retrieval accuracy and answer relevance.
- **Tech Stack:** Python, LangChain, Gemini API, Pinecone, Vector Search, RAG, Semantic Search, Prompt Engineering.

Publications

- Augmented Reality and Sustainability: Goals and Challenges | LinkTaylor & Francis (CRC Press) – 12/2025
- Author of Chapter 7 in a published book exploring the intersection of augmented reality and sustainability. The chapter discusses emerging technologies, practical applications, and long-term challenges in building sustainable digital ecosystems. Contributed research, technical analysis, and collaborative writing as part of an international academic publication.

Education

Vellore Institute of Technology <i>Bachelor of Technology in Computer Science</i>	07/2021 – 07/2025 <i>CGPA: 8.59</i>
• Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Computer Networks, Design & Analysis of Algorithms, Data Science	
St. Mary’s English High School, Jamshedpur <i>Class XII (Senior Secondary)</i>	03/2019 – 06/2020 <i>83%</i>
St. Mary’s English High School, Jamshedpur <i>Class X (Secondary)</i>	04/2017 – 05/2018 <i>89%</i>

Certifications

- Google Developers – Focused on applied AI/ML and real-world problem solving. | Certificate
- AWS Certified Solutions Architect – Credential demonstrating knowledge of designing scalable, secure, and distributed systems on AWS. | Certificate
- Data Science, Artificial Intelligence & Machine Learning – Skillians. Comprehensive training in data science methodologies, AI concepts, and machine learning techniques. | Certificate
- The Ultimate Job Ready Data Science Course – CodeWithHarry | Python, SQL, Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow, Machine Learning, Deep Learning, EDA, Feature Engineering.